

Part D: Failure and Recovery Study

COL7524 – Advanced Computer Networks

1. Link Failure and Timeline

We chose the direct link AS1 → AS4 to fail because it was the link that carried traffic just before the failure. In the Part B topology, the direct one-hop AS1 → AS4 route was selected by ns-3 because it has the lower routing cost. So, the other path AS1 → AS2 → AS3 → AS4 was not being used before time (t=6) s. We purposely failed the active AS1–AS4 link at (t=6) s to force traffic to use the available alternate path. We verify this transition in the packet trace; packets used the 192.168.4.1 interface before the failure and the 192.168.1.1 interface after the failure.

Interval	Event	Traffic Path
0–6 s	Normal operation	AS1 → AS4
6–9 s	AS1–AS4 failure	AS1 → AS2 → AS3 → AS4
9–13 s	Link restored	AS1 → AS4

2. FlowMonitor Results

Metric	Before (0–6 s)	During (6–9 s)	After (9–13 s)
Packets delivered	20	15	20
Packets lost	0	0	0
Average delay	2.4336 ms	11.8067 ms	2.4336 ms
Jitter	0 ms	0 ms	0 ms

3. Analysis

Prior to the failure, traffic was routed via the direct AS1 → AS4 path, with an average delay of 2.4336 ms. The link failed at (t=6) s, and the traffic was successfully rerouted through AS1 → AS2 → AS3 → AS4. No packet loss occurred, but the average delay increased to 11.8067 ms due to the longer three-hop path. The traffic then resumed its direct route, and the delay fell to 2.4336 ms once the AS1 → AS4 link was restored at t=9 s. So the experiment shows successful rerouting and recovery without packet loss.